

Validating stochastic tumor establishment near the B-cell fold

The old SDE “success” rate was a numerical artifact

Earlier scripts could stop when any state became negative, then classify the incomplete path from its last stored tumor value.

That allowed a failed solve to be labeled as a successful tumor.

Early stop + $T > 1$
= “success”

Retcode unchecked

Day 365 not required

Non-tumor failure counted

Day 365 or explicit
 $T < 1$ removal

Termination reason logged

Invalid paths excluded

Every seed auditable

The rebuilt pipeline validates definitions, numerics, and scale

1 · Criterion

2 · Solver

3 · Regression

Define the event

Verify trajectories

Recover the paper

Paper: T never falls
below 1 during 365 days.
A stricter criterion is
reported separately.

Zero noise, small noise,
seed repeatability,
nonnegativity, and time
resolution.

Match the published
four-state probability
before testing the five-
state extension.

All six validation gates passed before the near-fold pilot

The corrected implementation reproduces the original stochastic benchmark and removes hidden invalid trajectories.

Unit tests: 26/26 passed.

0

invalid paths
in the final pilot

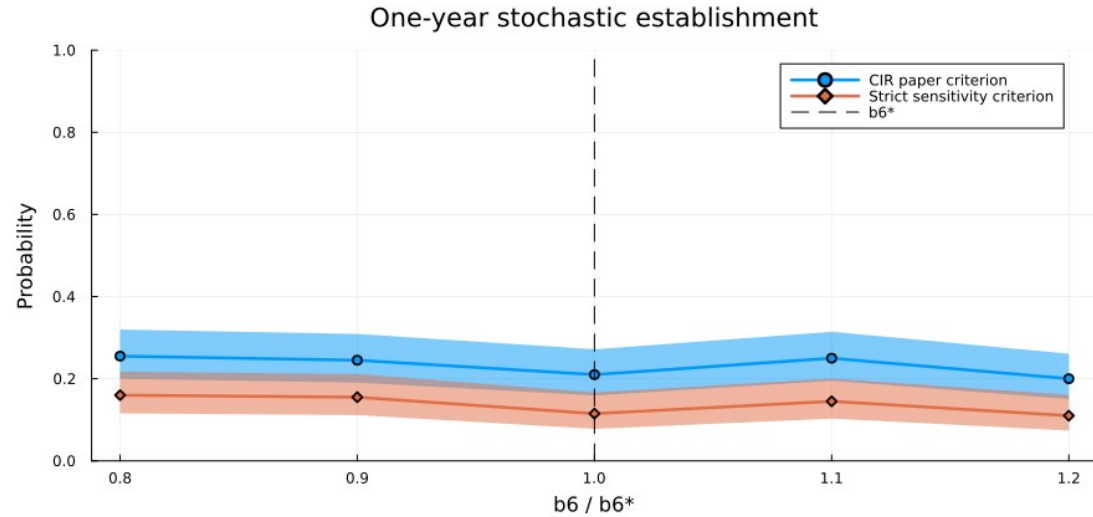
0.330

paper regression
published: 0.3217

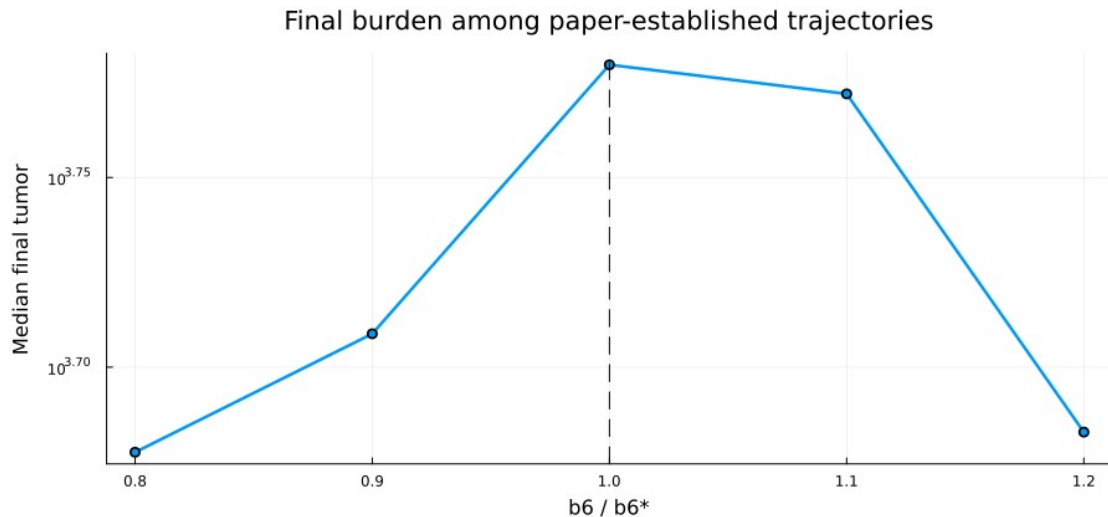
0.0395

small-noise mean error
acceptance: < 0.10

The pilot does not resolve a stochastic threshold near b_6^*



All 95% confidence intervals overlap, and the estimated curve is non-monotonic.



0.200–0.255

paper criterion

0.110–0.160

strict criterion

DECISION

Keep the saddle-node as the
primary result.

Choose: establishment or
true switching.

Then design the next test.